



LINUX PART OF THE SESSION

BASSANT MOHAMMED AMIN Practical Session

TASK 1

creating the folder and the files inside

```
bassant_ghaith@BassantGhaith: ~  
File Edit Tabs Help  
bassant_ghaith@BassantGhaith:~ $ cd~  
bash: cd~: command not found  
bassant_ghaith@BassantGhaith:~ $ pwd  
/home/bassant_ghaith  
bassant_ghaith@BassantGhaith:~ $ mkdir -p ~/reaction/{scripts,data,logs}  
bassant_ghaith@BassantGhaith:~ $ ls -R ~/reaction  
/home/bassant_ghaith/reaction:  
data logs scripts  
  
/home/bassant_ghaith/reaction/data:  
  
/home/bassant_ghaith/reaction/logs:  
  
/home/bassant_ghaith/reaction/scripts:  
bassant_ghaith@BassantGhaith:~ $
```

TASK 2

team file create / edit / show

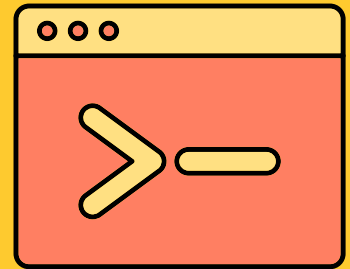
```
GNU nano 8.4 /home/bassant_ghaith/  
The only member name : Bassant Mohammed Amin Ali Ghaith  
Project : Reaction Time Game Scenario : 14
```

```
bassant_ghaith@BassantGhaith:~ $ touch ~/reaction/team.txt  
bassant_ghaith@BassantGhaith:~ $ nano ~/reaction/team.txt
```

```
bassant_ghaith@BassantGhaith:~ $ cat ~/reaction/team.txt  
The only member name : Bassant Mohammed Amin Ali Ghaith  
Project : Reaction Time Game Scenario : 14  
Date : 4 - 9 - 2026  
  
bassant_ghaith@BassantGhaith:~ $
```

TASK 3

Permission Control



```
bassant_ghaith@BassantGhaith:~ $ ls -l ~/reaction
total 16
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 data
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 logs
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 scripts
-rw-rw-r-- 1 bassant_ghaith bassant_ghaith 124 Sep  4 09:38 team.txt
bassant_ghaith@BassantGhaith:~ $ chmod 750 ~/reaction/team.txt
bassant_ghaith@BassantGhaith:~ $ nano ~/reaction/team.txt
bassant_ghaith@BassantGhaith:~ $ ls -l ~/reaction
total 16
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 data
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 logs
drwxrwxr-x 2 bassant_ghaith bassant_ghaith 4096 Sep  4 09:28 scripts
-rwxr-x--- 1 bassant_ghaith bassant_ghaith 237 Sep  4 09:45 team.txt
bassant_ghaith@BassantGhaith:~ $
```

TASK 4

temp reading

```
bassant_ghaith@BassantGhaith:~ $ nano ~/reaction/scripts/read_temp.py
bassant_ghaith@BassantGhaith:~ $ cat ~/reaction/scripts/read_temp.py
#!/usr/bin/env python3

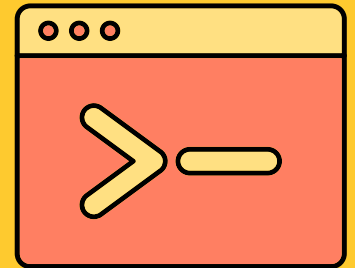
from datetime import datetime

with open("/sys/class/thermal/therman_zone0/temp") as f:
    value = int(f.read())//1000
now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
print(f"{now} Temp: {value}C")
```

```
bassant_ghaith@BassantGhaith:~ $ nano ~/reaction/scripts/read_temp.py
bassant_ghaith@BassantGhaith:~ $ python3 ~/reaction/scripts/read_temp.py
2026-09-04 10:21:56 Temp: 52C
bassant_ghaith@BassantGhaith:~ $ python3 ~/reaction/scripts/read_temp.py
2026-09-04 10:22:13 Temp: 52C
bassant_ghaith@BassantGhaith:~ $ python3 ~/reaction/scripts/read_temp.py
2026-09-04 10:22:22 Temp: 53C
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction
bassant_ghaith@BassantGhaith:~ $
```

TASK 5

Errors file



```
bassant_ghaith@BassantGhaith:~ $ cat ~/reaction/logs/errors.txt
```

Traceback (most recent call last):

File "<string>", line 1, in <module>

open('/this/path/does/not/exist')

~~^^

FileNotFoundError: [Errno 2] No such file or directory:

'/this/path/does/not/exist'

TASK 6

Sorting

```
bassant_ghaith@BassantGhaith:~ $ grep -o 'Temp: [0-9]*' ~/reaction/data/readings.txt | cut -d' ' -f2 | sort
/sorted.txt
bassant_ghaith@BassantGhaith:~ $ cat ~/reaction/data/sorted.txt
51
51
52
52
52
bassant_ghaith@BassantGhaith:~ $ head -1 ~/reaction/data/sorted.txt
tail -1 ~/reaction/data/sorted.txt
51
52
```

TASK 7

Soft Link creation

```
52
bassant_ghaith@BassantGhaith:~ $ ln -s ~/reaction/data/readings.txt ~/latest
bassant_ghaith@BassantGhaith:~ $ ls -l ~/latest
lrwxrwxrwx 1 bassant_ghaith bassant_ghaith 47 Sep  4 10:55 /home/bassant_ghaith/latest -> /home/bassant_ghaith/reaction/data/readings.txt
bassant_ghaith@BassantGhaith:~ $ cat ~/latest
2026-09-04 10:26:23 Temp: 52C
2026-09-04 10:26:38 Temp: 51C
2026-09-04 10:26:44 Temp: 52C
2026-09-04 10:26:50 Temp: 52C
2026-09-04 10:27:04 Temp: 51C
bassant_ghaith@BassantGhaith:~ $ ~/reaction/scripts/read_temp.py >> ~/reaction/data/readings.txt
bassant_ghaith@BassantGhaith:~ $ cat ~/latest
2026-09-04 10:26:23 Temp: 52C
2026-09-04 10:26:38 Temp: 51C
2026-09-04 10:26:44 Temp: 52C
2026-09-04 10:26:50 Temp: 52C
2026-09-04 10:27:04 Temp: 51C
2026-09-04 10:57:01 Temp: 51C
bassant_ghaith@BassantGhaith:~ $
```